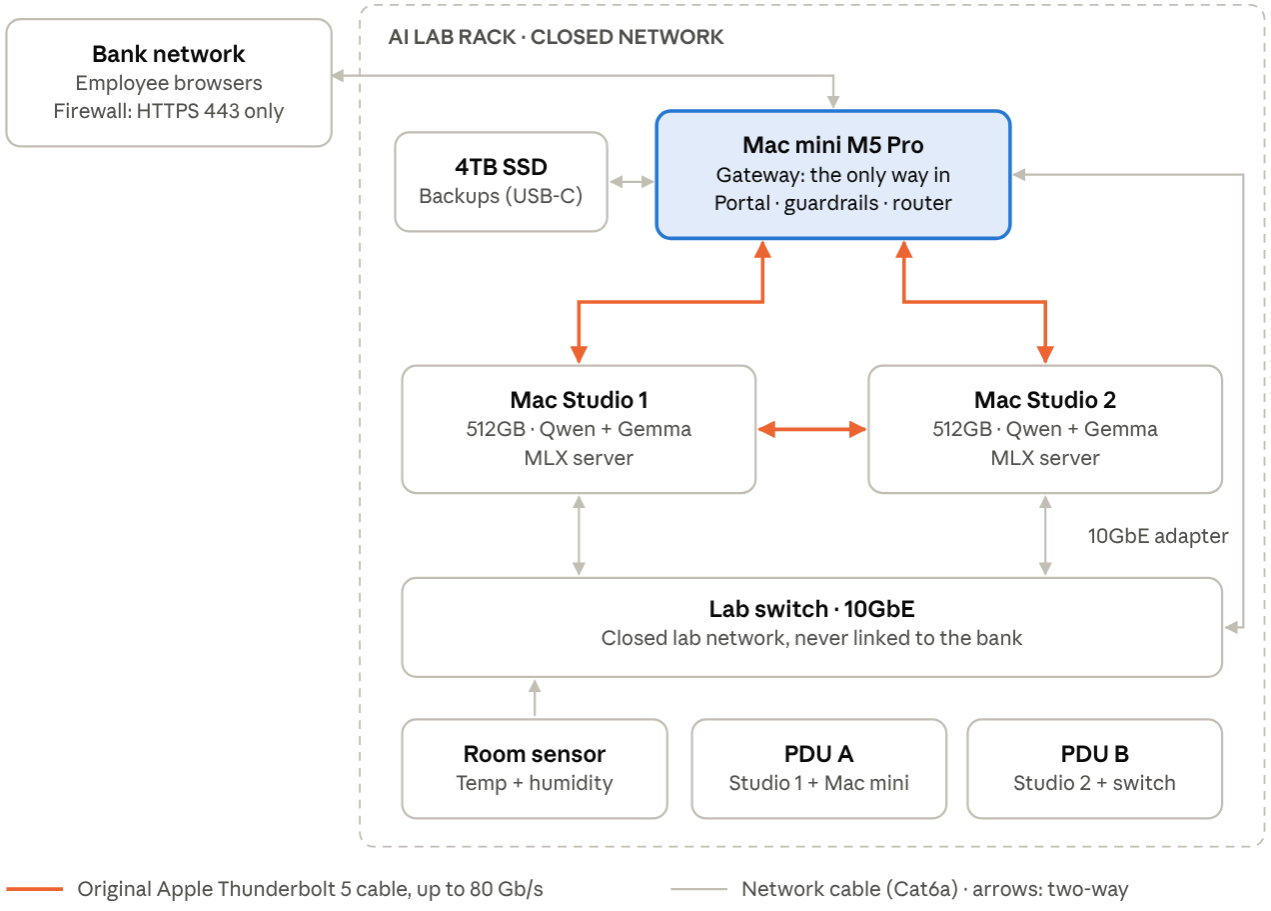
AI Lab Hardware & Site Plan

Oct 13, 2026 · @Hrishikesh

At a glance

Three Macs in one rack, joined by original Apple Thunderbolt 5 cables. Only the Mac mini connects to the bank network.

Only the Mac mini touches the bank network



AI Lab hardware · 3 Macs, 1 switch, 2 power circuits

How the rack will look

One 12U cabinet, with room for two more Mac Studios later. Front and rear views; reference only, not to scale.

AI Lab rack · front view

Illustration for reference · not to exact scale



Mac mini M5 Pro
gateway · the only way in

Mac Studio 1
M5 Ultra · 512GB

PDU A + PDU B
two 16A circuits

12U lockable cabinet
perforated doors · front door hidden

10GbE lab switch
U10 · ports face the rear

Room sensor
temperature + humidity

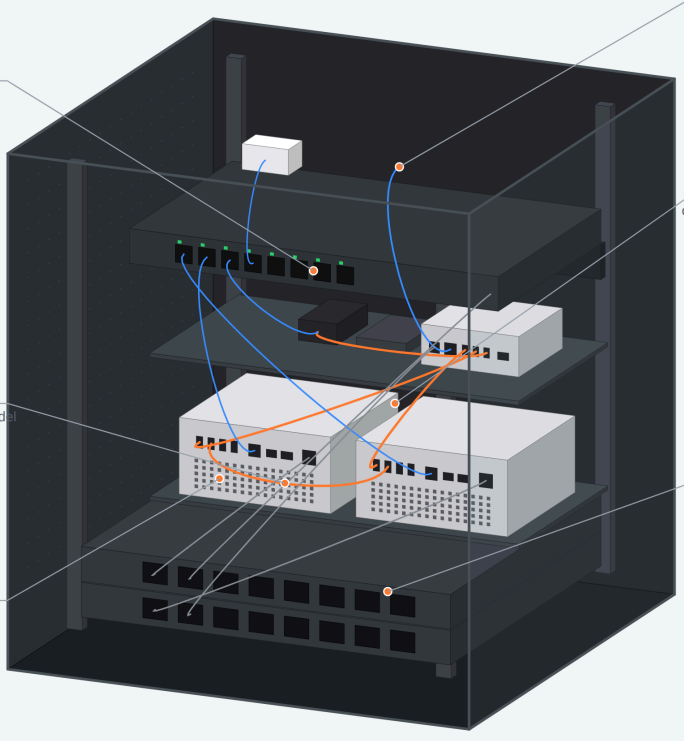
4TB encrypted SSD
backups, USB-C

Thunderbolt-to-10GbE adapter
backup lab link

Mac Studio 2
M5 Ultra · 512GB

AI Lab rack · rear view and cabling

Illustration for reference · not to exact scale



Switch ports (rear)
backup lab network · never linked to bank

Studio 1 ↔ Studio 2
Thunderbolt 5 · lets both run one large model

Rear vents
keep 10 cm clear behind each Mac

To bank network port
Cat6a from Mac mini · HTTPS 443 only

Thunderbolt 5 cables
original Apple, 1 m · Mac mini to each Studio

Power cords
original Apple cords to PDU A and PDU B

Thunderbolt 5 (Apple) · Cat6a network · Power

Hardware list

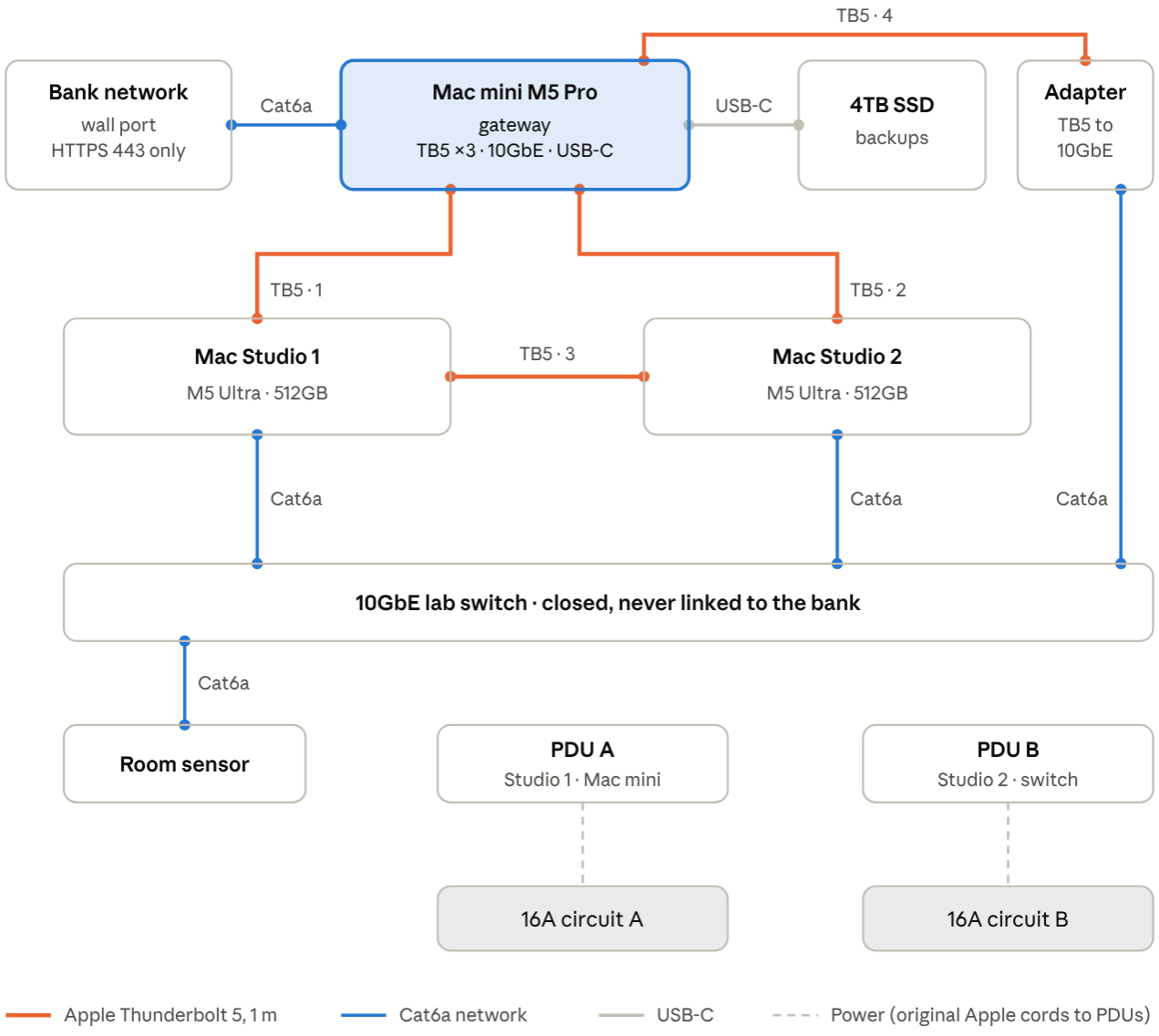
$39.5K in total, under the $40K ceiling. Planning prices (₹88 = $1), to be replaced by a reseller quote.

Item	Spec	Qty	Total
Mac Studio	M5 Ultra, 36-core CPU, 80-core GPU, 512GB, 2TB	2	$34.0K
Mac mini (gateway)	M5 Pro, 24GB, 512GB, 10GbE option	1	$2.5K
Apple Thunderbolt 5 Pro cable, 1 m	Original Apple	4 (1 spare)	$0.32K
Thunderbolt-to-10GbE adapter	Sonnet or OWC	1	$0.25K
10GbE switch	8-port managed, e.g. Cisco CBS350-8XT	1	$0.7K
Cat6a patch cables	5 × 1 m, 1 × 3 m	6	$0.05K
Encrypted SSD	4TB, USB-C	1	$0.4K
Rack PDU	16A, metered, surge-protected	2	$0.4K
Rack shelves	For 2 Studios and the Mac mini	1 set	$0.8K
Room sensor, cable labels and ties	Temperature + humidity	1 set	$0.1K
Total			**$39.5K**

Only if not already available: a 12U lockable cabinet ($0.5K) and two 16A circuits ($0.3K), usually from Facilities.

Connections

Every cable in the AI Lab, port to port



Wiring diagram · 4 Thunderbolt 5, 5 Cat6a, 1 USB-C, 2 power circuits

From	To	Cable
Mac mini	Mac Studio 1 and 2	Apple Thunderbolt 5: main link
Mac Studio 1	Mac Studio 2	Apple Thunderbolt 5: run large models together
Mac mini, both Studios, sensor	Switch	Cat6a: backup link, never connected to the bank
Mac mini	Bank network port	Cat6a: the only link to the bank, HTTPS 443

From	To	Cable
Mac mini	SSD	USB-C

Thunderbolt cables are 1 m, so the Macs sit on neighbouring shelves. The switch is not strictly needed for three Macs; we keep it as a backup path and for adding Macs later.

Power, room and security

- **Power:** about 1.0 kW at peak (385 W per Mac Studio, per Apple), 0.5–0.6 kW normally.
- **Circuits:** two 16A circuits with earthing, one PDU each. Studio 1 and Mac mini on A, Studio 2 and switch on B.
- **Cords:** original Apple power cords, PDU outlets to match. No UPS; the Macs restart automatically after a power cut.
- **Room:** 22–24°C (never above 27°C), 40–60% humidity, central AC on 24×7. Keep 10 cm clear behind each Mac.
- **Security:** about 10U in a lockable cabinet, in an access-controlled room. FileVault on; Wi-Fi and Bluetooth off.

Needed from other teams

Team	Need
Network and InfoSec	One bank network port; firewall allows HTTPS 443 to the Mac mini only; fixed IP, internal name and certificate
Identity and Endpoint	Active Directory sign-in; all three Macs in device management
InfoSec	An approved route to bring in models and updates, since the lab has no internet
Facilities	Rack space, two 16A circuits, room at 22–24°C around the clock
Procurement	Reseller quote and delivery date for the 512GB Mac Studio (in stores late October)